Akoizo Original Practice - Form 1

Division B | Disease Detectives | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which measure concerns new cases?

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

A. Incidence

B. Point prevalence

C. Mean disease duration

D. Reporting delay

Answer: _____

2. (2 points) RR for the example?

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

A. 4

B. 6

C. 0.3

D. 20

Answer: _____

3. (2 points) Which criterion creates circularity when testing a food exposure?

A. Requiring that food consumption to be a case

B. A defined onset interval

C. A standardized symptom threshold

D. A documented event population

Answer: _____

4. (2 points) Ten cases in 20 people versus ten in 200 means…

A. Different proportions

B. Equal risk necessarily

C. No denominator is needed

D. The second group has higher risk

Answer: _____

5. (2 points) Sensitivity's denominator is…

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

A. All reference-diseased people

B. All test-positive people

C. All reference-negative people

D. Only true positives

Answer: _____

6. (2 points) A more sensitive broad definition may…

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

A. Include more unrelated illness

B. Eliminate every false positive

C. Always improve causal inference

D. Guarantee laboratory confirmation

Answer: _____

7. (2 points) Reported cases may overrepresent…

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

A. Severe or readily tested illness

B. Every infection equally

C. Only uninfected people

D. A fixed fraction of every population

Answer: _____

8. (2 points) OR for the example?

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

A. 6

B. 4

C. 0.4

D. 0.1

Answer: _____

9. (2 points) Missing onset date should be…

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

A. Recorded as unknown or pending

B. Invented from the report date

C. Automatically coded no illness

D. Always ignored

Answer: _____

10. (4 points) Calculate 10/100 as an outbreak proportion.

Answer: _____
_____
_____

11. (4 points) Compute the risk difference.

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

Answer: _____
_____
_____

12. (4 points) Write the example's logic.

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

Answer: _____
_____
_____

13. (4 points) Why can prevalence rise without increased incidence?

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

Answer: _____
_____
_____

14. (4 points) Explain why OR differs from RR here.

Unit: Two-by-two tables and quantitative measures.
Worked-case data: a = 20, b = 30, c = 5, d = 45 in an exposure-outcome cohort.

Answer: _____
_____
_____

15. (4 points) Why version a changed definition?

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

Answer: _____
_____
_____

16. (4 points) Distinguish infection and reporting dates.

Unit: Epidemiological background and natural history.
Worked-case data: In a closed group of 100 initially well people, 10 develop the defined illness during one week. Separately, 15 of 100 residents have an existing chronic condition on survey day.

Answer: _____
_____
_____

17. (4 points) State RR = 4 in context.

Answer: _____
_____
_____

18. (4 points) How can two reviewers test clarity?

Unit: Constructing and applying case definitions.
Worked-case data: A synthetic case requires event attendance, vomiting or at least three loose stools, and onset on days 1-3. One attendee has one loose stool on day 2.

Answer: _____
_____
_____